

A2

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DS 120 Pop Quiz 1

September 2025

Instructions

- Unless otherwise instructed by a Teaching Fellow, the time limit for this quiz is **10 minutes**. Please manage your time accordingly.
- At the end of the quiz, please **submit** your completed paper to your Teaching Fellow.
- Only tools required for writing your answers are allowed on the desk. All other items must be put away.

Problem 1

1. Fill in the following table.

p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \vee q$	$\neg p \wedge \neg q$	$\neg p \vee \neg q$	$p \Rightarrow q$	$q \Rightarrow p$	$\neg p \vee q$
T	F	F	T	F	T	F	T	F	T	F
F	T	T	F	F	T	F	T	T	F	T

2. Identify if the following statements are true using the truth table above.

Note: Extend the truth table if necessary.

(a) $(\neg p \vee q) \Leftrightarrow (q \Rightarrow p)$

False

(b) $(p \wedge \neg q) \Leftrightarrow (p \vee \neg q)$

False

(c) $(\neg p \wedge q) \Leftrightarrow (\neg p \vee q)$

False

Problem 2

Determine the truth value of each expression below. Assume that all variables p, q, r, s, t represent logical propositions. Evaluate each expression step by step, using logical equivalences where appropriate. (Hint: For Part (2), you may find it helpful to refer back to the results from (1).)

(1) Assume that $p \rightarrow q$ is true.

1. $\neg(p \rightarrow q)$ False

2. $p \wedge \neg q$ False

3. $\neg(p \wedge \neg q)$ True

4. $\neg p \vee q$ True

F q
T T
~~W F~~
~~W F~~
F T

(2) Assume that $r \rightarrow (s \rightarrow t)$ is true.

1. $r \rightarrow (\neg s \vee t)$ T

2. $\neg r \vee (\neg s \vee t)$ F

3. $(\neg r \vee \neg s) \vee t$ F

4. $\neg(r \wedge s) \vee t$ T

5. $(r \wedge s) \rightarrow t$ F

T T
F T
F F